AyuAware Today Section

Technical Product Requirements Document

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1. Executive Summary

What We're Building

The Today section is the **daily home base** of AyuAware — the screen users see every time they open the app. It combines three essential functions:

1. **At-a-glance health metrics** — CHI score, dosha balance, environmental exposure
2. **Conversational AI guidance** — The Digital Twin provides personalized insights
3. **Quick actions** — Easy access to common tasks like logging meals or starting practices

Unlike typical health dashboards that overwhelm users with charts and numbers, the Today section leads with conversation. The Twin greets the user, summarizes what's important, and invites dialogue. Metrics are visible but secondary — supporting context rather than the main event.

Why It Matters

This is where the relationship between user and Twin becomes real. During onboarding, users met their Twin. In the Today section, they **live with it** daily. The Twin:

- Notices patterns the user might miss
- Connects dots across sleep, activity, nutrition, and environment
- Speaks in accessible Ayurvedic language
- Adapts its guidance to time of day and current state

Success means users open the app not to check numbers, but to **hear what their Twin has to say**.

Key Differentiator

Most health apps present data and leave interpretation to the user. AyuAware's Today section does the opposite — the Twin interprets data and presents **insights**. The user can drill into data if curious, but they never have to.

2. Section Overview

2.1 The Screen Structure

```
┌──────────────
```

```
│
│
│   [🪷] Twin's morning insight message...
│
│
│        User's question...                                [User]
│
│
│   [🪷] Twin's response with embedded metric card...
│
│
│   [🪷] Quick action grid...
│
│
│
│
├──────────────────────────────────────────────────────────────────────────────────────┤
│                                  SUGGESTED QUESTIONS
│
│   💭 Why is my CHI low today?
│
│   💭 What's the best time for my practice?
│
├──────────────────────────────────────────────────────────────────────────────────────┤
│                                    CHAT INPUT BAR
│
│   📷   Ask your twin anything...                                                  🎤
│        │
└──────────────────────────────────────────────────────────────────────────────────────┘
```

2.2 Information Hierarchy

Priority	Element	Purpose
1	Twin's insight message	Primary value — personalized guidance
2	Compact metrics	Quick health snapshot
3	Chat input	Always accessible interaction

4	Suggested questions	Lower barrier to engagement
5	Quick actions	Task shortcuts
6	Header navigation	System functions

2.3 Design Philosophy

Chat-First, Data-Second The conversation scroll area dominates the screen. Metrics are compact and above the fold, but they're context for the conversation, not the main attraction.

Time-Aware Intelligence Everything adapts to when the user opens the app:

- Morning: Focus on overnight recovery, day planning
- Afternoon: Energy check, meal guidance
- Evening: Wind-down recommendations, reflection
- Night: Sleep preparation, gentle reminders

Progressive Disclosure

- Compact metrics show just the essentials
- Tap any card to explore full details
- Twin explains metrics conversationally when relevant
- Users never need to hunt for information

Persistent Access The chat input is always visible and reachable. No matter where the user is scrolling in their conversation history, they can always ask a question.

3. User Journey Breakdown

3.1 Morning Open (Primary Use Case)

Context: User wakes up, opens AyuAware as part of their morning routine.

What They See:

```
Good morning, Sarah 🌅
Friday, August 11, 2023

┌──────────┐  ┌──────────┐  ┌──────────┐
│   CHI    │  │   NSL    │  │ Exposure │
│   72%    │  │   82%    │  │ ☀️ 72°   │
│ Kapha↑   │  │ Vata↑    │  │          │
└──────────┘  └──────────┘  └──────────┘
```

Twin's Morning Message:

"Good morning! I've analyzed your overnight data. Here's what I see:

Your recovery looks good overall — steady HRV and balanced heart rate. However, I'm noticing a slight Kapha accumulation from lower activity levels yesterday.

For today, I'd suggest starting with some light movement to get energy flowing. Even a 10-minute walk would help balance things out."

Why This Works:

- Greeting is warm and personal
- Metrics give quick context without requiring analysis
- Twin proactively explains what matters
- Actionable suggestion included
- User can respond, ask questions, or just absorb and go

3.2 Midday Check-In

Context: User opens app during lunch break, curious about energy dip.

What They See:

```
Good afternoon, Sarah ☀️
Friday, August 11, 2023

┌──────────┐  ┌──────────┐  ┌──────────┐
│   CHI    │  │   NSL    │  │ Exposure │
│   68%    │  │   78%    │  │ ☀️ 85°   │
│ Pitta↑   │  │ Pitta↑   │  │          │
└──────────┘  └──────────┘  └──────────┘
```

User Types: "I'm feeling sluggish after lunch"

Twin Responds:

"That makes sense given what I'm seeing. Your Pitta has been elevated since late morning — likely from the heat and possibly a heavier lunch.

A few things that might help right now:

- Cool water with a squeeze of lime
- A few minutes away from screens
- If possible, a short walk in shade

Your digestion typically peaks around 1-2pm (Pitta time), so feeling heavier afterward is natural. Would you like me to suggest some cooling foods for dinner?"

Why This Works:

- Validates user's experience ("That makes sense")
- Connects symptoms to Ayurvedic patterns
- Offers immediate, practical suggestions
- Opens door to further conversation

3.3 Evening Wind-Down

Context: User opens app after dinner, thinking about sleep.

What They See:

```
Good evening, Sarah 🌆
Friday, August 11, 2023

┌──────────┐  ┌──────────┐  ┌──────────┐
│   CHI    │  │   NSL    │  │ Exposure │
│   74%    │  │   80%    │  │ 🌙 68°   │
│  Vata↑   │  │  Vata↑   │  │          │
└──────────┘  └──────────┘  └──────────┘
```

Twin's Evening Message:

"Winding down for the evening? I notice your Vata has picked up since this afternoon — this sometimes happens with a busy day.

To help settle into good sleep, you might try:

- A warm drink (chamomile or warm milk with nutmeg)
- Gentle stretching or legs-up-the-wall
- Dimming lights in the next hour

Based on your patterns, sleeping before 10:30pm tends to give you the best recovery. You're currently on track! 🌙"

Why This Works:

- Acknowledges end of day
- Notes current state without alarming
- Suggestions fit the moment
- References user's personal patterns
- Encouraging without being preachy

3.4 Exploring a Metric

Context: User taps on the CHI card to understand it better.

Transition: Card expands with a hero animation into a detailed view (separate screen/modal).

Detailed CHI View Contains:

- Full circular visualization with breakdown
- Trend over past 7 days
- Contributing factors (HRV, sleep, activity, etc.)
- Twin explanation of what's affecting the score
- Related protocols or recommendations

Return Path: Swipe down or tap back to return to Today view. Conversation continues where they left off.

3.5 Quick Actions Flow

Context: Twin presents a quick action grid after morning insight.

Twin Message:

"What would you like to do?"

```
┌──────────────┐ ┌──────────────┐ │ 🥗 │ │ 🏃 │ │ Log │ │ Start │ │Breakfast │ │
Workout │ └──────────────┘ └──────────────┘

┌──────────────┐ ┌──────────────┐ │ 🧘 │ │ 📋 │ │ Meditate │ │ View │ │ │
│Protocols │ └──────────────┘ └──────────────┘
```

User Taps "Log Breakfast":

1. Action card shows tap feedback
2. Nutrition logging flow opens (modal or navigation)
3. After logging, user returns to Today
4. Twin acknowledges: "Great! I've noted your breakfast. I'll factor that into your afternoon guidance."

Why This Works:

- Actions are contextual to time of day
- Presented as conversation, not toolbar
- Completion acknowledged by Twin
- Seamless return to main flow

3.6 Suggested Question Flow

Context: User sees suggested questions below the chat.

Suggested Questions:

```
💭 Why is my CHI low today?
💭 What's the best time for my practice?
💭 How is the moon affecting me?
```

User Taps "Why is my CHI low today?":

1. Question chip fades out
2. Question appears as user message immediately
3. Typing indicator shows
4. Twin responds with detailed explanation

Twin Response:

"Your CHI is at 68% today, which is lower than your usual 75-80%. Here's what's contributing:

Sleep: You got 6.2 hours vs your ideal 7-8. HRV was also lower than usual.

Activity: Yesterday was quite sedentary — only 3,200 steps vs your average of 7,000.

Stress signals: Your heart rate variability suggests elevated stress in the afternoon.

The good news: these are all adjustable factors. Would you like some specific suggestions for bringing your CHI back up?"

Why This Works:

- Lowers barrier to engagement
- Questions are contextually relevant
- Feels like natural conversation
- Twin provides substantive answers

3.7 Historical View

Context: User wants to see what their Twin said yesterday.

How to Access:

- Scroll up in chat to see previous messages
- Or tap the date to open date selector

Date Selector Appears:

```
←    August 10, Thursday    →
```

Historical Mode Indicators:

- Warm background tint on date selector
- Metrics show historical values
- Chat shows that day's conversation
- "Today" button appears to return

Why This Matters: Users can review:

- What the Twin noticed on a specific day
- Their questions and the Twin's answers
- How metrics have changed over time
- Patterns they discussed previously

4. Technical Architecture

4.1 High-Level Architecture

```
┌──────────────
```

```
│   │   │  Header    │ │  Metrics   │ │   Chat     │ │   Input
│   │   │
│   │   │  Bar       │ │  Cards     │ │  Messages  │ │   Bar
│   │   │
│   │   └────────────┘ └────────────┘ └────────────┘
│   │
└───────────────┘   │   │
│
└──────────────────────────────────────────────────────────────────────────────
─┘    │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                          │
                                          ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                                STATE LAYER
│
│   ┌──────────────────┐   ┌──────────────────┐
┌──────────────────┐
│   │   Chat Store     │   │  Metrics Store   │   │    UI Store
│   │
│   │                  │   │                  │   │
│   │
│   │ • Messages       │   │ • CHI data       │   │ • Selected
date │      │
│   │ • Typing state   │   │ • NSL data       │   │ • Loading
state │      │
│   │ • Suggestions    │   │ • Exposure       │   │ • Error state
│   │
│   │ • Quick actions  │   │ • Trends         │   │ • Input focus
│   │
│   └──────────────────┘   └──────────────────┘
└──────────────────┘       │
└──────────────────────────────────────────────────────────────────────────────
──────────┘
                                          │
                                          ▼
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                               SERVICE LAYER
│
│   ┌──────────────────────────┐
┌──────────────────────────────────────┐     │
│   │    Today Agent           │   │      Health Data Service
│   │
```

```
│   │     (Claude API)      │   │
│   │                        │   │  • Fetch metrics from backend
│   │  • Morning insights    │   │  • Calculate derived values
│   │  • Response gen        │   │  • Aggregate wearable data
│   │  • Context awareness   │   │  • Cache management
│   └────────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────┘
                                             │
                                             ▼
┌──────────────────────────────────────────────────────────────────────────────┐
│                                DATA LAYER
│
│   ┌────────────────────────┐
┌──────────────────────────────────────┐     │
│   │   Local Storage        │   │      Backend APIs
│   │                        │   │
│   │  • Chat history        │   │  • Chat history sync
│   │  • Cached metrics      │   │  • Metrics aggregation
│   │  • User preferences    │   │  • Claude API proxy
│   │  • Offline queue       │   │  • Real-time WebSocket
│   └────────────────────────┘
└──────────────────────────────────────┘     │
└──────────────────────────────────────────────────────────────────────────────┘
```

4.2 Data Flow: Opening the App

```
User opens app
        │
        ▼
┌──────────────────┐
│ Load cached data │ ── Immediate display of last-known
metrics
```

```
└──────────────────────┘      and recent messages
          │
          ▼
┌──────────────────────┐
│ Fetch fresh data     │ ── Parallel requests:
└──────────────────────┘      • GET /metrics/summary
          │                   • GET /chat/history (last 24h)
          ▼                   • GET /chat/suggestions
┌──────────────────────┐
│ Check for new        │ ── Has Twin generated morning insight?
│ proactive message    │      (Server-side scheduled job)
└──────────────────────┘
          │
          ├── Yes ──► Display Twin's proactive message
          │
          └── No ──► Generate morning insight via Claude API
                              │
                              ▼
                     Display generated message
```

4.3 Data Flow: User Sends Message

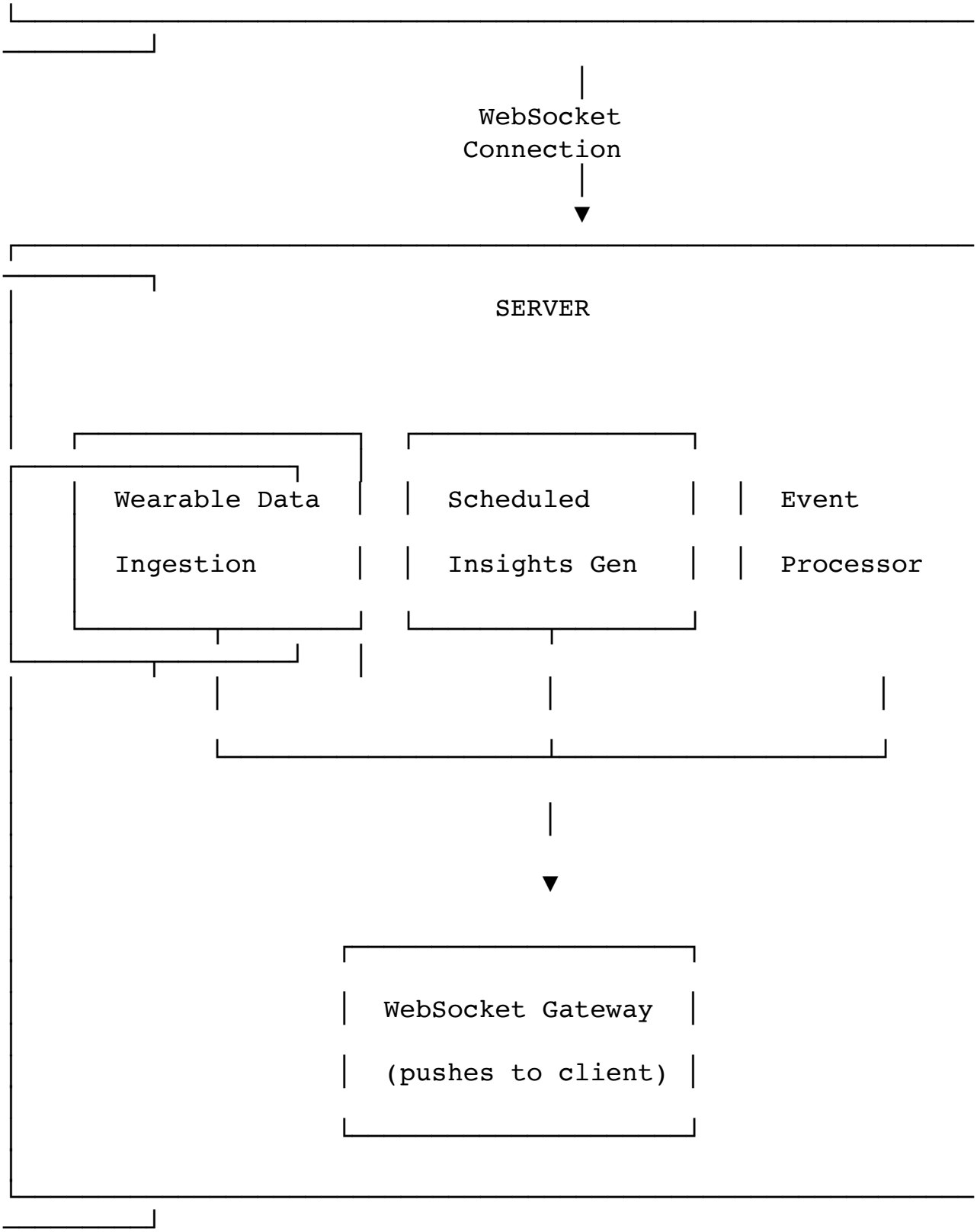
```
User types and sends message
          │
          ▼
┌──────────────────────┐
│ Optimistic update    │ ── Message appears immediately
(pending state)
└──────────────────────┘
          │
          ▼
┌──────────────────────┐
│ Show typing          │ ── Twin typing indicator
│ indicator            │
└──────────────────────┘
          │
          ▼
┌──────────────────────┐
│ Build context        │ ── Gather:
│ for Claude           │      • User's constitution
└──────────────────────┘      • Current metrics
          │                   • Recent conversation
          ▼                   • Time of day
                              • Any anomalies
┌──────────────────────┐
```

```
│ Call Claude API    │ ── POST to backend (which calls
Anthropic)
└────────────────────┘
        │
        ▼
┌────────────────────┐
│ Process response   │ ── Parse message content
└────────────────────┘    Check for rich content (metrics,
actions)
        │                 Extract suggested follow-ups
        ▼
┌────────────────────┐
│ Update UI          │ ── Hide typing indicator
└────────────────────┘    Display Twin's response
                          Update suggested questions
                          Scroll to bottom
```

4.4 Real-Time Architecture

For live updates (e.g., new wearable data, background insights), we use WebSockets:

```
┌──────────────────────────────────────────────────────────────────────────────
──────────┐
│                          CLIENT (Mobile App)
│
│
│
```



5. AI Integration & Prompting Strategy

5.1 The Role of Claude in Today Section

The Today section is where Claude does its heaviest lifting. Unlike onboarding (which is semi-scripted), the Today section requires **dynamic, contextual intelligence**:

Capability	Description
Morning Insights	Analyze overnight data, proactively share what's notable
Conversational Response	Answer questions with Ayurvedic context
Metric Interpretation	Explain numbers in accessible terms
Recommendation Generation	Suggest actions based on current state
Pattern Recognition	Notice trends across days/weeks
Contextual Awareness	Adapt tone and content to time of day

5.2 System Prompt for Today Twin

```
You are the Ayurvedic Digital Twin for AyuAware, serving as
the user's daily
wellness companion. You have access to their health data,
understand their
constitution (Prakriti), and observe their current state
(Vikriti).

## YOUR ROLE

You are NOT a health dashboard or data reporter. You are a
wise companion who:
- Notices what's meaningful in the data
- Explains patterns in accessible terms
- Connects symptoms to Ayurvedic understanding
- Offers practical, actionable guidance
- Celebrates progress and acknowledges challenges

## YOUR KNOWLEDGE

You understand:
- The three doshas (Vata, Pitta, Kapha) and their qualities
- Daily rhythms (Dinacharya) and how time affects the body
- Seasonal influences (Ritucharya) on constitution
- The concept of Agni (digestive fire) and its importance
- How modern metrics (HRV, sleep stages, etc.) relate to
Ayurvedic concepts

## COMMUNICATION STYLE

Tone: Warm, insightful, supportive — like a knowledgeable
friend
Length: Concise but complete — typically 2-4 short paragraphs
```

Language: Accessible first, Sanskrit terms introduced naturally with context
Perspective: "I notice..." rather than "Your data shows..."

WHAT YOU DO

DO:
- Lead with the most relevant insight
- Connect multiple data points into a coherent picture
- Offer 2-3 practical suggestions when appropriate
- Acknowledge the user's experience when they share how they feel
- Ask clarifying questions when helpful
- Celebrate positive patterns and progress

DO NOT:
- List all available metrics
- Use medical terminology or make diagnoses
- Overwhelm with too many suggestions at once
- Be preachy or prescriptive
- Ignore what the user explicitly says they're feeling
- Repeat the same advice without variation

TIME-OF-DAY AWARENESS

Morning (5 AM - 12 PM):
- Focus on overnight recovery
- Set intentions for the day
- Recommend morning practices
- Lighter, energizing tone

Afternoon (12 PM - 5 PM):
- Check on energy and digestion
- Address mid-day dips
- Pitta-time awareness
- Practical, focused tone

Evening (5 PM - 9 PM):
- Wind-down preparation
- Reflect on the day
- Sleep optimization
- Calming, settling tone

Night (9 PM - 5 AM):
- Sleep encouragement

```
- Minimal engagement
- Gentle, quiet tone

## HANDLING SPECIFIC SCENARIOS

When user reports feeling unwell:
- Validate their experience first
- Look for correlating data patterns
- Offer comfort-focused suggestions
- Don't minimize their experience

When metrics show concerning patterns:
- Mention it conversationally, not alarmingly
- Provide context ("This sometimes happens when...")
- Suggest gentle adjustments
- Recommend professional consultation if truly concerning

When user asks about something outside your scope:
- Acknowledge the question kindly
- Explain your limitations briefly
- Suggest appropriate resources
- Redirect to what you CAN help with

## IMPORTANT BOUNDARIES

- Never diagnose medical conditions
- Never recommend stopping medications
- Never provide emergency medical advice
- Always suggest professional help for serious concerns
- Remember you're a wellness companion, not a doctor
```

5.3 Context Building for Each Request

Every time the user sends a message (or we generate a proactive insight), we build a context package:

```
CONTEXT PACKAGE STRUCTURE:

## User Profile
- Name: {user_name}
- Constitution: {prakriti} (e.g., "Vata-Pitta")
- Current imbalance tendency: {vikriti_tendency}
- Member since: {join_date}
- Preferences: {communication_preferences}

## Current Metrics (Today)
- CHI Score: {chi_value}% ({chi_status})
```

```
- CHI Components:
  - HRV: {hrv_value} ({hrv_trend})
  - Sleep: {sleep_hours}h ({sleep_quality})
  - Activity: {steps} steps ({activity_level})
  - Recovery: {recovery_score}%

- NSL Balance:
  - Vata: {vata_percentage}%
  - Pitta: {pitta_percentage}%
  - Kapha: {kapha_percentage}%
  - Dominant imbalance: {dominant_imbalance}

- Exposure:
  - Weather: {weather_condition}, {temperature}
  - Moon phase: {moon_phase}
  - Season: {season}
  - Location: {location}

## Recent Patterns (7-day view)
- CHI trend: {chi_trend} (e.g., "declining over 3 days")
- Sleep pattern: {sleep_pattern}
- Notable events: {notable_events}

## Conversation Context
- Time of day: {current_time} ({time_period}: morning/
afternoon/evening/night)
- Last interaction: {last_interaction_time}
- Recent topics discussed: {recent_topics}
- Previous 5 messages: {message_history}

## Today's Activities (if any logged)
- Meals: {logged_meals}
- Practices: {logged_practices}
- Symptoms: {reported_symptoms}
```

5.4 Prompt Templates by Scenario

Morning Insight Generation:

```
Generate a morning insight message for {user_name}.

{CONTEXT_PACKAGE}

Your message should:
1. Greet them appropriately for the morning
```

2. Summarize the most notable insight from their overnight data
3. Connect it to their constitution if relevant
4. Offer 1-2 practical suggestions for the day
5. End with an opening for conversation (optional question or invitation)

Keep it warm and concise — 2-3 short paragraphs maximum.
Do not list every metric. Focus on what's most meaningful.

Response to User Question:

The user asked: "{user_message}"

{CONTEXT_PACKAGE}

Respond to their question with:
1. Direct acknowledgment of what they asked
2. Relevant information from their data if applicable
3. Ayurvedic context that helps them understand
4. Practical suggestions if appropriate
5. Optional follow-up question if it would help

Remember:
- Validate their experience if they're sharing how they feel
- Don't lecture — have a conversation
- If their question is outside your scope, redirect kindly

Metric Explanation (when user taps "Why is X low?"):

The user wants to understand why their {metric_name} is at {metric_value}.

{CONTEXT_PACKAGE}

Explain by:
1. Acknowledging the specific value and how it compares to their normal
2. Identifying 2-3 likely contributing factors from their data
3. Connecting to Ayurvedic concepts where natural
4. Offering perspective (is this concerning or normal variation?)
5. Suggesting what might help, if appropriate

Use headers sparingly. Keep it conversational.

Quick Action Suggestion:

```
Based on the current context, suggest relevant quick actions.

{CONTEXT_PACKAGE}

Select 4 actions most relevant to:
- Time of day: {time_period}
- User's current state: {current_state_summary}
- What they haven't done yet today: {pending_activities}
- Their typical patterns: {usual_activities}

Return as JSON:
{
  "question": "What would you like to do?",
  "actions": [
    { "id": "action_id", "icon": "emoji", "label": "Action
Label" },
    ...
  ]
}
```

5.5 Suggested Questions Generation

The suggested questions should be contextually relevant, not generic. We generate them based on:

```
Generate 3-4 suggested questions for the user to ask.

{CONTEXT_PACKAGE}

Questions should be:
1. Relevant to their CURRENT state and data
2. Things they might genuinely wonder about
3. Varied in type (explain metric, get advice, understand
pattern, etc.)
4. Answerable by you with the available data

Examples of GOOD contextual questions:
- "Why is my CHI lower than yesterday?" (if CHI dropped)
- "What's causing my Kapha to rise?" (if Kapha elevated)
- "How can I improve my sleep tonight?" (if sleep was poor)
- "Is my digestion affecting my energy?" (if both are
suboptimal)

Examples of BAD generic questions:
- "What is Ayurveda?"
- "Tell me about my constitution"
- "How does the app work?"
```

```
Return as JSON array:
["question1", "question2", "question3"]
```

5.6 Rich Content Detection

The Twin can include embedded content in its responses. The backend parses Claude's response for special markers:

```
When you want to include a metric visualization, use:
[METRIC:chi] or [METRIC:nsl] or [METRIC:exposure]

When you want to offer quick actions, use:
[ACTIONS]
- Log Breakfast|🥗|log_meal_breakfast
- Start Workout|🏃|start_workout
- Meditate|🧘|start_meditation
[/ACTIONS]

When you want to highlight a key insight, use:
[HIGHLIGHT]
Your HRV has improved 15% this week — a sign your
nervous system is responding well to the evening
routine changes.
[/HIGHLIGHT]

These markers will be parsed and rendered as rich components
in the app.
```

5.7 Conversation Memory Strategy

For coherent conversations, we maintain context:

Short-term (Current Session):

- Full message history since app opened
- All messages included in Claude context

Medium-term (Past 7 Days):

- Summary of key topics discussed
- Significant insights shared
- User questions and concerns raised

Long-term (Persistent):

- User's stated preferences
- Recurring patterns noted
- Goals they've mentioned
- Things they've said they liked/disliked

```
MEMORY MANAGEMENT:
```

```
Before each Claude call:
1. Include last 10 messages verbatim
2. Include 7-day topic summary
3. Include relevant long-term preferences

After notable conversations:
1. Extract key topics (via Claude)
2. Store in medium-term memory
3. Update long-term preferences if explicit

This allows the Twin to reference:
- "As we discussed yesterday..."
- "I remember you mentioned..."
- "Since you prefer X over Y..."
```

6. Data Models & State Management

6.1 Core Data Types

Metrics:

```
CHI (Constitutional Health Index):
- value: number (0-100)
- status: 'excellent' | 'good' | 'moderate' | 'low'
- trend: 'improving' | 'stable' | 'declining'
- components:
  - hrv: { value, trend, status }
  - sleep: { hours, quality, trend }
  - activity: { steps, activeMinutes, trend }
  - recovery: { score, trend }
- contributingFactors: string[] (what's affecting it)

NSL (Natural State Level - Dosha Balance):
- vata: number (0-100)
- pitta: number (0-100)
- kapha: number (0-100)
- dominantImbalance: 'vata' | 'pitta' | 'kapha' | 'balanced'
- trend: 'improving' | 'stable' | 'worsening'

Exposure:
- weather:
  - condition: string
  - temperature: number
  - humidity: number
```

```
  - uvIndex: number
- moon:
  - phase: string
  - illumination: number
- season: string
- location: { city, region }
```

Messages:

```
Message:
- id: string (unique identifier)
- type: 'twin' | 'user' | 'system'
- content: string (text content)
- timestamp: Date
- status: 'pending' | 'sent' | 'delivered' | 'failed'
- richContent?: RichContent (optional embedded components)
- metadata?: {
    generatedBy?: 'proactive' | 'response'
    contextUsed?: string[]
    suggestions?: string[]
  }

RichContent:
- type: 'metric' | 'action-grid' | 'highlight' | 'image'
- data: varies by type
  - metric: { metricType: 'chi' | 'nsl' | 'exposure',
snapshot: MetricData }
  - action-grid: { question: string, actions: Action[] }
  - highlight: { text: string }
  - image: { uri: string, caption?: string }

Action:
- id: string
- icon: string (emoji or icon name)
- label: string
- route: string (navigation target)
- metadata?: Record<string, any>
```

Suggested Questions:

```
SuggestedQuestion:
- id: string
- text: string
- context: string (why this question is relevant)
- priority: number (ordering)
- expiresAt?: Date (time-sensitive questions)
```

6.2 State Structure

```
TodayState:
  # Date Context
  - selectedDate: Date (today or historical)
  - isHistorical: boolean

  # Metrics
  - metrics:
    - chi: CHIMetric | null
    - nsl: NSLMetric | null
    - exposure: ExposureMetric | null
    - lastUpdated: Date
    - isLoading: boolean
    - error: Error | null

  # Chat
  - messages: Message[]
  - isLoadingMessages: boolean
  - hasMoreMessages: boolean (for pagination)
  - typingIndicator: boolean

  # Suggestions
  - suggestedQuestions: SuggestedQuestion[]
  - quickActions: Action[]

  # Input
  - inputText: string
  - inputMode: 'text' | 'voice'
  - isRecording: boolean
  - attachedImage: ImageAttachment | null

  # UI
  - isRefreshing: boolean
  - scrollPosition: number
  - keyboardVisible: boolean
```

6.3 State Management Approach

We use **Zustand** with **separate stores** for different concerns:

```
Stores:

1. chatStore
   - Messages array
   - Typing state
   - Send/receive actions
   - Optimistic updates
```

```
   - WebSocket listeners

2. metricsStore
   - CHI, NSL, Exposure data
   - Fetch and cache logic
   - Historical data
   - Trend calculations

3. todayUIStore
   - Selected date
   - Input state
   - Scroll position
   - Modal states
   - Loading indicators

Benefits:
- Clear separation of concerns
- Independent updates (metrics don't re-render chat)
- Easier testing
- Simpler debugging
```

6.4 Caching Strategy

Data Type	Cache Duration	Invalidation
Metrics	5 minutes	New data from wearable, manual refresh
Chat history	Persistent (local DB)	Sync on app open
Suggestions	15 minutes	After user asks a question
Quick actions	30 minutes	Time of day changes
User profile	Session	Profile update

7. Component Requirements

7.1 Header Bar

Purpose: Navigation and quick access to settings/notifications

Elements:

- Menu icon (hamburger) — Opens side navigation
- Title ("Today") — Static
- User name — Shows current user
- Notification bell — Shows unread count badge
- Settings gear — Opens settings

Behaviors:

- Sticky at top of screen
- Subtle shadow when content scrolls beneath
- Notification badge pulses when new notifications

7.2 Time-Based Greeting

Purpose: Personal welcome that sets context

Elements:

- Greeting text with contextual emoji
- Current date

Time Variations:

Time	Greeting	Emoji	Subtle Gradient
5 AM - 12 PM	Good morning, {Name}	🌅	Warm yellow tint
12 PM - 5 PM	Good afternoon, {Name}	☀️	Bright orange tint
5 PM - 9 PM	Good evening, {Name}	🌆	Cool purple tint
9 PM - 5 AM	Good evening, {Name}	🌙	Dark blue tint

7.3 Compact Metric Cards

Purpose: At-a-glance health snapshot

Layout: 3 equal-width cards in a row

Card Contents:

Card	Label	Primary Value	Secondary Info
CHI	"CHI"	Percentage (72%)	Trend indicator + dominant factor
NSL	"NSL"	Percentage (82%)	Dominant dosha with arrow
Exposure	"Exposure"	Weather + temp icons	Current conditions

Interactions:

- Tap any card → Navigate to detailed view
- Tap feedback: scale down, border highlight
- Pull-to-refresh updates all cards

States:

- Loading: Shimmer placeholder
- Error: Show "--" with "Tap to retry"
- Stale: Show data with subtle indicator

7.4 Chat Message Components

Twin Message:

- Left-aligned with avatar
- Light background (#F8FAFB)
- Chat tail on top-left corner
- Timestamp below
- Can contain rich content (metrics, actions)

User Message:

- Right-aligned, no avatar
- Teal background (#E8F4F3)
- Chat tail on top-right corner
- Timestamp below
- Pending state: slightly faded

System Message:

- Centered
- No bubble
- Decorative lines on sides
- Used for milestones, check-in confirmations

Rich Content (Embedded in Twin Messages):

- Metric Card: Shows visualization with action button
- Action Grid: 2x2 grid of tappable actions
- Highlight: Emphasized text block

7.5 Typing Indicator

Purpose: Show Twin is "thinking"

Design:

- Same styling as Twin message bubble
- Three animated dots
- Wave animation (dots bounce sequentially)
- Appears with Twin avatar

Timing:

- Show after 300ms delay from user send
- Hide when response arrives
- If response takes >10s, show "Taking longer than usual..." text

7.6 Suggested Questions

Purpose: Lower barrier to engagement

Layout:

- Vertical list of pill-shaped buttons
- Above the input bar

- Collapsible/dismissible

Behavior:

- Tap question → It becomes user message
- Questions update based on context
- Fade out on tap, response appears

7.7 Chat Input Bar

Purpose: Always-accessible message input

Elements:

- Camera button (left) — Attach image
- Text input field (center) — Expandable
- Voice button (right) — Hold to record
- Send button (far right) — Active when input has content

States:

- Default: Placeholder text, send disabled
- Typing: Text visible, send enabled
- Voice: Recording UI, waveform, timer
- Image attached: Thumbnail preview

Sticky Behavior:

- Fixed at bottom
- Keyboard pushes it up
- Safe area padding at bottom

7.8 Date Selector (Historical Mode)

Purpose: Navigate to past days

Design:

- Appears below greeting when viewing history
- Warm background tint indicates "past"
- Left/right arrows for navigation
- "Today" badge when viewing current day

Behavior:

- Changing date loads that day's metrics and messages
- Smooth transition between days
- Can't navigate to future

8. API Requirements

8.1 REST Endpoints

Get Today's Summary:

```
GET /api/v1/today/summary?date={YYYY-MM-DD}

Response:
{
  "date": "2025-01-31",
  "greeting": {
    "text": "Good morning, Sarah",
    "emoji": "🌅",
    "period": "morning"
  },
  "metrics": {
    "chi": { ... },
    "nsl": { ... },
    "exposure": { ... }
  },
  "suggestedQuestions": [ ... ],
  "quickActions": [ ... ],
  "hasUnreadNotifications": true
}
```

Get Chat History:

```
GET /api/v1/chat/history?date={YYYY-MM-DD}
&limit=20&before={message_id}

Response:
{
  "messages": [ ... ],
  "hasMore": true,
  "oldestMessageId": "msg_xxx"
}
```

Send Message:

```
POST /api/v1/chat/message

Request:
{
  "content": "Why is my CHI low today?",
  "attachments": []
}

Response:
{
  "userMessage": { ... },
  "twinResponse": { ... },
  "updatedSuggestions": [ ... ]
}
```

Get Metric Details:

```
GET /api/v1/metrics/{metric_type}?date={YYYY-MM-DD}

Response:
{
  "metric": { ... full metric data ... },
  "trend": { ... 7-day trend ... },
  "insights": [ ... relevant insights ... ],
  "relatedProtocols": [ ... ]
}
```

Log Quick Action:

```
POST /api/v1/actions/log

Request:
{
  "actionId": "log_meal_breakfast",
  "data": { ... action-specific data ... }
}

Response:
{
  "success": true,
  "twinAcknowledgment": "Great! I've noted your breakfast..."
}
```

8.2 WebSocket Events

Server → Client:

```
metrics.updated
{
  "type": "chi" | "nsl" | "exposure",
  "data": { ... updated metric ... },
  "reason": "wearable_sync" | "manual_entry" |
"scheduled_refresh"
}

twin.message
{
  "message": { ... proactive message from Twin ... },
  "reason": "morning_insight" | "anomaly_detected" |
"reminder"
}

suggestions.updated
```

```
{
  "questions": [ ... new suggested questions ... ],
  "reason": "context_change" | "time_based"
}
```

Client → Server:

```
chat.typing
{
  "isTyping": true
}

chat.read
{
  "lastReadMessageId": "msg_xxx"
}
```

8.3 Claude API Integration

Endpoint: Backend proxies to Anthropic Messages API

Request Flow:

1. Client sends message to backend
2. Backend builds context package
3. Backend calls Claude with system prompt + context + user message
4. Backend parses response for rich content markers
5. Backend returns structured response to client

Error Handling:

- Timeout (15s) → Return fallback message
- Rate limit → Queue and retry with backoff
- Content filter → Return gentle deflection

9. Success Metrics

9.1 Engagement Metrics

Metric	Target	Measurement
Daily Active Users	Track growth	Users who open Today section
Messages per session	> 2	Average messages sent per session
Session duration	> 3 minutes	Time spent in Today section
Return rate (next day)	> 40%	Users who return within 24h
Suggested question tap rate	> 20%	Taps on suggestions / impressions

9.2 Quality Metrics

Metric	Target	Measurement
Response satisfaction	> 80%	Thumbs up / total responses (if implemented)
Conversation depth	> 3 turns	Average turns per conversation
Feature discovery	Track	Users who tap metric cards
Quick action usage	Track	Actions completed from grid

9.3 Technical Metrics

Metric	Target	Measurement
Initial load time	< 2s	Time to first meaningful content
Message send to response	< 3s	End-to-end latency
Metric refresh time	< 1s	Time to update after pull-to-refresh
Crash-free sessions	> 99.5%	Sessions without crashes
Offline capability	Graceful	Cached content shown when offline

9.4 Health Metrics (Business)

Metric	Target	Measurement
Insights read rate	> 70%	Morning insights viewed
Metric exploration	> 30%	Users who tap into detailed views
Protocol engagement	Track	Users who start protocols from Today
Retention correlation	Positive	Today usage vs 30-day retention

10. Implementation Considerations

10.1 Performance Optimization

Initial Load Strategy:

1. Show cached metrics immediately (from last session)
2. Show cached last 5 messages immediately
3. Fetch fresh data in background
4. Animate updates when fresh data arrives

Message List Performance:

- Virtualized list (only render visible messages)
- Pagination for history (load 20 at a time)
- Image lazy loading with placeholders
- Reuse message bubble components

Claude Response Streaming: Consider streaming Claude's response for longer messages:

- Show typing indicator initially
- Stream text as it arrives
- Better perceived performance
- Requires WebSocket or SSE support

10.2 Offline Handling

When Offline:

- Show cached metrics with "Last updated X ago" indicator
- Allow reading chat history
- Queue outgoing messages for send when online
- Show offline banner at top

When Back Online:

- Sync queued messages
- Fetch fresh metrics
- Check for proactive messages missed
- Subtle notification of refresh

10.3 Error Handling

Network Errors:

- Retry with exponential backoff
- Show inline error with retry button
- Never lose user's typed message

Claude API Errors:

- Timeout: "I'm taking longer than usual to think about this..."
- Rate limit: Queue and notify user of delay
- Content filter: "I'm not sure how to help with that. Could you rephrase?"

Data Errors:

- Invalid metric data: Show "--" with refresh option
- Missing messages: Show gap indicator, offer to refresh

10.4 Accessibility Requirements

Screen Reader:

- Message roles announced: "Message from your Twin" / "Your message"
- Metrics announced: "CHI at 72 percent, status good, Kapha elevated"
- Actions have clear labels
- Typing indicator announced

Motor Accessibility:

- All touch targets ≥ 44x44px
- Swipe alternatives for all gestures

- Voice input as alternative to typing

Visual Accessibility:

- All text meets WCAG AA contrast
- Don't rely on color alone for status
- Support Dynamic Type scaling
- Reduce motion option respected

10.5 Analytics Events

Track These Events:

Event	Properties	Purpose
`today_opened`	source, time_of_day	Track entry points
`metric_card_tapped`	metric_type	Feature discovery
`message_sent`	has_attachment, length	Engagement depth
`suggested_question_tapped`	question_id	Suggestion relevance
`quick_action_tapped`	action_id	Action usefulness
`historical_date_viewed`	days_ago	History usage
`pull_to_refresh`	—	Manual refresh behavior
`session_ended`	duration, messages_sent	Session quality

10.6 Testing Priorities

Unit Tests:

- Message rendering (all types)
- Metric calculations and formatting
- Time-of-day logic
- Rich content parsing

Integration Tests:

- Full send/receive message flow
- Metric fetch and display
- Date navigation
- Offline/online transitions

E2E Tests:

- Morning flow: Open app → See insight → Ask question → Get response
- Historical view: Navigate to yesterday → See past conversation
- Quick action: Tap action → Complete flow → See acknowledgment

Manual QA:

- Twin personality consistency
- Timing and animations feel natural
- Edge cases (very long messages, rapid inputs)

- Different times of day

Appendix A: Message Examples

Morning Insight (Good Day)

"Good morning! I've been looking at your overnight data, and I'm seeing some really nice patterns.

Your HRV stayed steady through the night, and you got 7.5 hours of quality sleep. Your body seems well-rested and ready for the day.

Today looks like a good day to challenge yourself a bit if you're feeling up for it. Your Vata is well-grounded, which means you have stability to work from. 🪷"

Morning Insight (Concerning Pattern)

"Good morning. I want to share something I'm noticing in your data.

Your CHI has been declining over the past three days — from 78% down to 65%. Looking at the patterns, it seems connected to shorter sleep and higher stress indicators in the evenings.

This isn't alarming, but it's worth paying attention to. Would you like to talk about what's been going on, or would some suggestions for tonight be helpful?"

Response to Symptom Report

User: "I've been feeling really tired all afternoon"

"That makes sense given what I'm seeing in your data today.

Your Kapha has been elevated since this morning, which often shows up as heaviness and low energy, especially in the afternoon. Combined with the humid weather (78% humidity today), it's creating conditions that naturally slow things down.

A few things that might help right now:

- A short, brisk walk — even 5 minutes can shift the energy
- Something slightly warming to drink (ginger tea, or just warm water with lemon)
- If you can, some light stretching or movement

The good news: this tends to lift as evening approaches. How are you feeling otherwise?"

Response to Specific Question

User: "Why is my NSL showing Pitta high?"

"Your Pitta is at 45% right now, which is about 10% higher than your baseline.

Looking at what might be contributing:

Heat exposure — It's been a hot week (averaging 88°F), and your outdoor activity has been higher than usual. Pitta accumulates with heat.

Meal timing — I notice you've been eating later in the evenings this week. Late digestion can aggravate Pitta, especially when it's heavy food.

Intensity — Your workouts have been more intense lately. Great for fitness, but it does generate internal heat.

None of this is problematic on its own, but together they're pushing Pitta up. Would you like some cooling practices to bring it back to balance?"

Appendix B: Quick Actions by Time of Day

Morning (5 AM - 12 PM)

Icon	Label	Action
🥗	Log Breakfast	Open meal logger
🧘	Morning Practice	Start morning routine
📝	Set Intention	Open journaling
💊	Log Supplements	Open supplement tracker

Afternoon (12 PM - 5 PM)

Icon	Label	Action
🥗	Log Lunch	Open meal logger
🏃	Start Workout	Open activity tracker
🧘	Quick Meditation	Start 5-min meditation
📋	View Protocols	Open active protocols

Evening (5 PM - 9 PM)

Icon	Label	Action
🥗	Log Dinner	Open meal logger
🌙	Wind Down	Start evening routine
📖	Reflect	Open evening journal
💤	Sleep Prep	View sleep recommendations

Night (9 PM - 5 AM)

Icon	Label	Action
🌙	Sleep Mode	Minimize notifications
📖	Quick Note	Simple note capture
💤	Can't Sleep?	Sleep support guidance
❓	Quick Question	Ask Twin

Appendix C: Metric Status Definitions

CHI Score Ranges

Range	Status	Color	Twin's Framing
80-100%	Excellent	#4A9B8E (teal)	"Your body is thriving"
60-79%	Good	#F8C471 (gold)	"Solid foundation today"
40-59%	Moderate	#E89B7F (coral)	"Room for some support"
0-39%	Low	#E89B7F (coral)	"Let's focus on recovery"

Dosha Balance Interpretation

Dominant	Status	Twin's Approach
Vata elevated	Movement, anxiety, dryness	Grounding, warming, calming suggestions
Pitta elevated	Heat, intensity, irritability	Cooling, moderating suggestions
Kapha elevated	Heaviness, sluggishness	Stimulating, lightening suggestions
Balanced	All within 10% of each other	Maintenance suggestions

End of Document